INCH-POUND
MIL-I-24092/8
4 May 1989

MILITARY SPECIFICATION SHEET

INSULATING RESIN, SOLVENTLESS, THIXOTROPIC, BAKING, SEMI-RIGID, FOR DIP PROCESSING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-I-24092.

REQUIREMENTS:

1. Grade - SHT: Clear, solventless, thixotropic, semi-rigid baking resin.

2. Class - 155 to 200. The cured resin shall meet the requirements of the specified thermal class when determined in accordance with MIL-I-24092.

3. Composition - IV. Composition IV consists of resins that are in solution with either an organic monomer or an organic diluent. Such materials are not normally considered solvents since they are reactive and become part of the cured polymer.

4. The resin shall meet the qualification and quality conformance requirements specified in tables I and II.

AMSC N/A FSC 5970
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TABLE I. Qualification requirements, grade SHT.

Characteristics (tests)	Units	Requirements
Catalyzed storage life	Months, minimum	12
Vapor pressure	mm Hg, maximum	2
Weight loss on cure	Average weight percent, maximum	30.0
Flash point	°C, minimum	93
Viscosity:		
at 2 r/min	Centipoise	15000
at 20 r/min	Centipoise	5000
Thixotropic index	Minimum	1.1
Build, as received	Mils, minimum	2.0
Drainage	Weight percent, maximum	25
Dielectric strength:		
96/23/50	Volts per mil, minimum	1500
96/23/96	Percent decrease from 96/23/50 value, maximum	25
24/23/water	Percent decrease from 96/23/50 value, maximum	25
Dissipation factor:		
at 23°C	Percent, maximum	0.10
at 150°C	Percent, maximum	5.5
Dielectric constant:		
at 23°C	Maximum	3.0
at 150°C	Maximum	4.5
Hardness	Shore D, minimum	85
Bond strength		
at 23°C	Pounds force, minimum	30
at 150°C	Pounds force, minimum	5

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TABLE I. Qualification requirements, grade SHT - Continued.

Characteristics (tests)	Units	Requirements
Thermal endurance:		
Class 155	Temperature index, °C, minimum	155
Class 180	Temperature index, °C, minimum	180
Class 200	Temperature index, °C, minimum	200
Salt water proofness	Hours, minimum	100
Hydrolytic stability	Percent decrease in bond strength, maximum	50
Chemical resistance:		
Hydraulic fluid	Percent decrease in bond strength, maximum	25
Lubrication oil		25
Cleaning fluid		25
Distilled water		25
Detergent solution		25

TABLE II. Quality conformance requirements, grade SHT.

Characteristics (tests)	Units	Requirements
Weight loss on cure	Average weight percent, maximum	30
Gel point	Percent of as-received value	± 10
Viscosity at 20 r/min	Centipoise maximum	15000
Build, as received	Mil, minimum	2.0
Specific gravity	Percent of as-received value	5.0
Hardness	Shore D, minimum	60
Bond strength at 23°C	Pounds force, minimum	30

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NOTE:

Part or Identifying Number (PIN). Part number shall be M24092/8-(dash number indicating thermal class). For example, M24092/8-155 describes a material meeting the requirements of this specification and having class 155 thermal endurance.

Custodians:
Army - ER
Navy - SH
Air Force - 20

Review activities:
Army - MI, AR
Navy - EC, YD, OS
Air Force - 99
DLA - GS, DS

User activities:
Army - ME
Navy - MC, AS

Preparing activity:
Navy - SH
(Project 5970-0661-08)

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